Response Parameter Sensitivity: Comparing Original and Revised pmsimstats Implementations

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Abstract

Figure 5 of Hendrickson et al. (2020) explores how the response trajectory parameters (Gompertz maximum and rate for each of three factors) affect statistical power to detect a biomarker–treatment interaction across four trial designs. This paper reproduces Figure 5 using both the original publication code and the revised code, comparing the results to assess how the methodological corrections (AR(1) autocorrelation, exponential correlation decay, `nlme::lme` with `corCAR1`) affect the sensitivity of power to response parameter choices. Both simulations use 200 replicates per cell.

Contents

1 Background **2**

2 Methodology **2**
2.1 Original Code (Commit `42ac030`) . . . 2
2.2 Revised Code (Current HEAD) . . . 2
2.3 Key Differences at Carryover = 0 . . . 2

3 Original Code: Figure 5 (Publication Parameters) **3**

4 Revised Code: Figure 5 (Corrected Parameters) **4**

5 Comparison and Analysis **4**
5.1 Qualitative Patterns Preserved . . . 4
5.2 Why Absolute Power Is Lower in the Revised Code . . . 5
5.3 Design Rankings . . . 5

6 Conclusions **6**

1 Background

Figure 5 of the publication presents two panels:

- **Panel A:** The effect of response parameter *maximum values* on power. The Gompertz `max` parameter for each factor (BR, ER, TR) is varied across $\{5, 8, 11\}$, producing $3^3 = 27$ combinations. Fixed at $N = 35$, $c_{bm} = 0.6$, carryover $= 0$. Note: `sd = max` in the publication's parameterization.
- **Panel B:** The effect of response parameter *rate values* on power. The Gompertz `rate` parameter is varied across $\{0.2, 0.35, 0.5\}$. Fixed at $N = 35$, $c_{bm} = 0.3$, carryover $= 0$.

These parameters control the *mean trajectory* of each factor and the between-person variability around it (since `sd = max` in Panel A). They are distinct from the correlation structure parameters analyzed in the Figure 4 comparison.

2 Methodology

2.1 Original Code (Commit `42ac030`)

- Compound symmetry autocorrelation ($\rho = 0.8$)
- Step-function BM–BR correlation (full c_{bm} when BR mean $\neq 0$)
- `lme4::lmer` analysis with random intercept
- $c_{bm} = 0.6$ (Panel A), $c_{bm} = 0.3$ (Panel B)
- No carryover ($t_{1/2} = 0$), no censoring

2.2 Revised Code (Current HEAD)

- AR(1) autocorrelation ($\rho = 0.7$, $\rho^{|t_i - t_j|}$)
- Exponential BM–BR correlation decay ($\lambda_{cor} = \ln 2 / t_{1/2}$, auto-computed; at carryover $= 0$, $\lambda_{cor} = 0$, equivalent to original behavior)
- `nlme::lme` with `corCAR1` residual correlation
- $c_{bm} = 0.5$ (Panel A), $c_{bm} = 0.25$ (Panel B) — constrained by PD feasibility
- No carryover, no censoring

2.3 Key Differences at Carryover = 0

At zero carryover, several of the DGP corrections have no effect:

- The correlation decay (λ_{cor}) is irrelevant because there are no off-drug timepoints with carryover residual (for designs with off-drug periods, the raw BR mean is exactly zero, so no correlation is assigned under either rule).
- The scale factor removal has no effect (no carryover decay to scale).
- The timepoint-1 fix applies only to on-drug timepoints, which behave the same under both rules at carryover $= 0$.

The differences that *do* matter at carryover $= 0$:

1. c_{bm} **value:** 0.6 vs 0.5 (Panel A), 0.3 vs 0.25 (Panel B). Lower c_{bm} reduces the interaction signal, lowering power.

2. **Autocorrelation:** CS $\rho = 0.8$ vs AR(1) $\rho = 0.7$. AR(1) produces lower correlations at distant timepoints, reducing the repeated-measures advantage.
3. **Analysis model:** `lmer` vs `nlme + corCAR1`. The `corCAR1` model correctly accounts for residual autocorrelation, producing wider (but honest) confidence intervals.

3 Original Code: Figure 5 (Publication Parameters)

`figure5_42ac030.png`

Figure 1: Figure 5 reproduced from the original commit `42ac030` (200 replicates). Panel A: $c_{bm} = 0.6$, CS $\rho = 0.8$, `lmer`. Panel B: $c_{bm} = 0.3$. Matches the publication's qualitative patterns.

4 Revised Code: Figure 5 (Corrected Parameters)

`figure5_HEAD.png`

Figure 2: Figure 5 from the revised code (200 replicates). Panel A: $c_{bm} = 0.5$, AR(1) $\rho = 0.7$, `nlme + corCAR1`. Panel B: $c_{bm} = 0.25$. Qualitative patterns preserved; absolute power is lower.

5 Comparison and Analysis

5.1 Qualitative Patterns Preserved

Both versions show the same qualitative sensitivity patterns:

1. **Higher BR max increases power.** The drug-specific signal grows with BR max, making the biomarker–treatment interaction easier to detect. This is the dominant effect in Panel A.

2. **Higher ER max decreases power.** The expectancy-related response confounds the drug-specific signal. Designs with open-label phases (OL, OL+BDC, N-of-1) are more affected because their expectancy values are higher.
3. **Higher TR max decreases power.** The time-variant (non-specific) response adds noise uncorrelated with the biomarker, increasing the residual variance and reducing the signal-to-noise ratio.
4. **Rate effects are less consistent.** Changes in the Gompertz rate parameters have design-dependent effects that do not follow a simple pattern. This matches the publication's finding and reflects the complex interaction between trajectory shape and the analysis model's ability to detect the interaction term.

5.2 Why Absolute Power Is Lower in the Revised Code

Three factors reduce power in the revised code, all of which are consequences of methodological corrections rather than deficiencies:

1. **Lower c_{bm}.** Panel A uses $c_{bm} = 0.5$ instead of 0.6 (constrained by PD feasibility). The biomarker–treatment interaction signal scales with c_{bm}, so a 17% reduction in c_{bm} produces a substantial power reduction, especially in the mid-range of the power curve.
2. **Correct standard errors via `corCAR1`.** The `nlme::lme` model with `corCAR1` estimates and accounts for residual autocorrelation, producing wider confidence intervals than `lmer` (which assumes independent residuals). The wider CIs are correct; the narrower CIs from `lmer` are anti-conservative, inflating both power and Type I error.
3. **AR(1) vs compound symmetry.** Under AR(1) at $\rho = 0.7$, distant timepoints have near-zero correlation ($0.7^{20} \approx 0$), while under CS at $\rho = 0.8$ all pairs maintain correlation 0.8. The reduced within-person consistency under AR(1) means each participant's repeated measures carry less redundant information, weakening the power advantage of crossover and N-of-1 designs.

The publication's higher power was partly genuine (higher c_{bm}) and partly artifact (underestimated standard errors from misspecified residual structure). The revised estimates are more conservative but statistically honest.

5.3 Design Rankings

The relative ranking of designs is preserved across both versions:

Table 1: Design power ranking (Panel A, BR max = 11, ER max = 5, TR max = 5)

Design	Original	Revised
N-of-1	Highest	Highest
CO	High	High
OL+BDC	Moderate	Moderate
OL	Lowest	Lowest

The design comparison—the primary purpose of the simulation—is robust to the methodological corrections.

6 Conclusions

1. The qualitative sensitivity patterns in Figure 5 are preserved under the revised code. Higher BR max increases power, higher ER/TR max decreases power, and rate effects are design-dependent.
2. The design ranking (N-of-1 > CO > OL+BDC > OL) is unchanged by the corrections.
3. Absolute power is lower due to three principled corrections: reduced c_{bm} (PD constraint), correct standard errors (`corCAR1`), and AR(1) autocorrelation. These produce honest power estimates with validated Type I error control.
4. The response parameter sensitivity analysis is independent of the carryover-related corrections (correlation decay, scale factor removal) because Figure 5 uses carryover = 0. The differences between the two versions are entirely attributable to the autocorrelation structure, the analysis model, and the c_{bm} constraint.

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